

Florida Residential Market Intelligence

Stage 2/5: Classification Pipeline

Group 3

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Preprocessing

For the first phase of this project, we performed EDA and used linear regression to explore the price prediction potential of our dataset for the real state market in Florida as reported in the previous document. Although the prediction errors for that baseline were still high, the analysis provided a useful structure in terms of **market intelligence**, and the best use for that first milestone within our long run project will be a **dashboard** with *price tendencies* and a clear *signal of the direction of the market*.

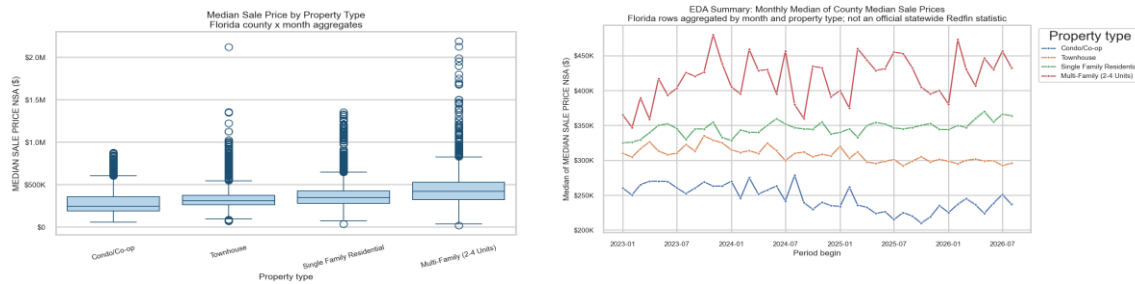


Figure 1. Market Level Screening for Dashboard

Evaluation Plan: Our initial classification question was: *Using this month's market conditions, can we predict whether the median sale price will increase next month?* We also aim to detect the counties and property types that will show a signal of price increment within the next month. After comparing three classification models some signal was detected, HistGradientBoosting captures better non linear patterns than Logistic Regression and DT and we confirmed that our dataset captures well enough *direction and market context*, not so much yet reliable for *property-level investment decisions*. A brief overview follows:

Pipeline: Data → Preprocessing (OneHotEncoder for categorical features + StandardScaler for Logistic Regression) → Classification Target → Chronological Train/Test Split → Logistic Regression / Decision Tree / HistGradientBoosting → Evaluation

Target: 1 = next month's median sale price is higher; 0 = equal or lower.

Models: Logistic Regression and Decision Tree as baselines, plus HistGradientBoosting as a stronger nonlinear model.

Evaluation Metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Validation Strategy: Train on earlier months, we left on purpose the month of December 2025 out as a *separation month*, to avoid data leakage, and test on January–July 2026. Time-aware cross-validation also keeps future months out of past training avoiding training on future data.

Model Comparison Approach: All three models use the same target, predictors, test period, and evaluation metrics.

At this stage we have not produced yet a production-ready investment model, but we did establish a clean classification baseline, identified the limits of the current data, and gave the project a clear direction for the next iteration.

Next Steps: Having a proper market screening gives us a good base to move forward in the direction of property level *investment decisions*. One of the business decisions moving forward is that we'll depart from the one month prediction *to test 3-, 6-, and 12-month horizons*, which better match the timeline of buying, renovating, and reselling a property.

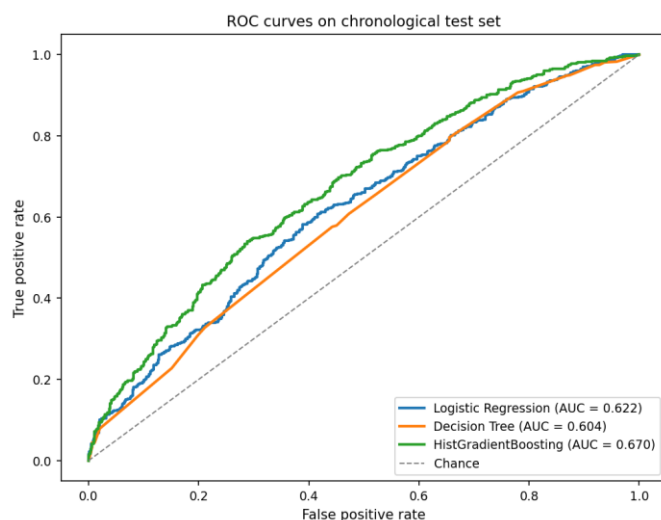
After evaluation of the metrics and comparing models we see that *it is necessary to look for more granular* ZIP-code, neighborhood, and eventually property-level *data*. That way we can create time-based features such as price momentum, inventory changes, trends in terms of day on market of properties, and rolling averages. We want to move toward a tool focused on property flips that considers expected resale value, renovation cost, holding costs, profit, and risk.

On the technical side we will continue into the planned *Neural Models Integration* stage and compare whether more flexible models capture additional signal.

Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV Accuracy Mean	CV F1 Mean	CV ROC-AUC Mean
Logistic Regression	0.597	0.598	0.582	0.590	0.622	0.575	0.538	0.616
Decision Tree	0.567	0.564	0.576	0.570	0.604	0.545	0.496	0.575
HistGradientBoosting	0.619	0.620	0.609	0.615	0.670	0.580	0.585	0.620

The models found some predictive signal in the county-level monthly data, but not enough to support a real investment decision by itself. The cross-validation results were consistent with the final test results, with HistGradientBoosting showing the strongest overall performance, it clearly shows the highest metrics. These classification results are useful as a baseline: they show that a stronger nonlinear model helps, but the current data and one-month horizon are still limited. As we see on the table, HistGradientBoosting performed best, but the improvement was still modest. The best accuracy was about 62% and ROC-AUC was 0.67.

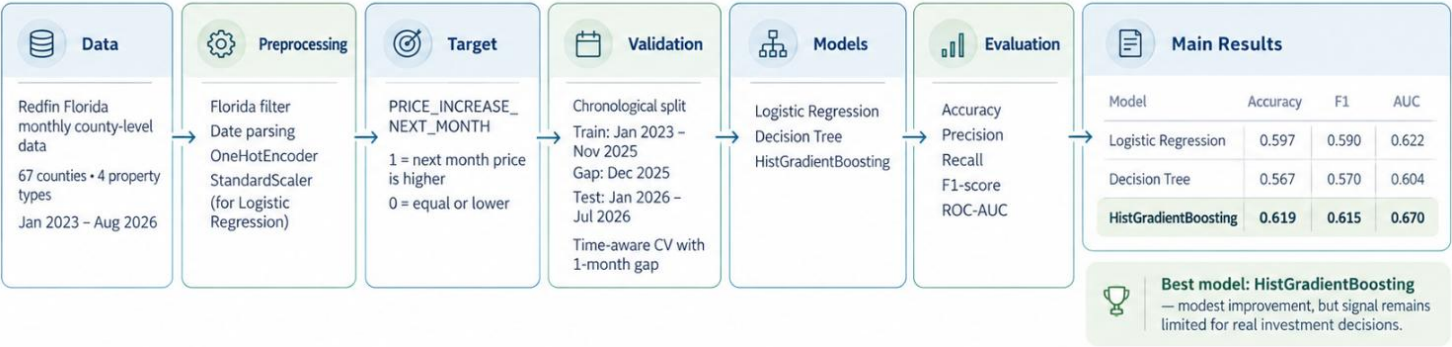


The long-term goal is not only to predict whether a county median will move next month. The project is intended to become more useful for real estate investment and potential flips. A one-month county-level signal is too broad and too short-term for that decision, so we concluded we will move to a 3, 6 and 12 month horizon and

granularize our data to property or at least Zip code level.

To integrate the **Artificial Intelligence** component, the final system will include a **RAG-based chatbot** that allows users to ask questions about the market data and model results, with the possibility of adding search and constraint-based property screening for *investment decisions*.

Pipeline Diagram



Data source: Redfin Housing Market Tracker. Florida monthly county-level data, January 2023–August 2026.